CMAS/PWS Manual Launch Guide

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Contents

Manual Launch Guide — CMAS/PWS Emergency Alert Lab **1**
Quick Reference . . . 1
Phase 1: Pre-flight Checks (2 min) . . . 2
Phase 2: Start Core Network (2 min) . . . 2
Phase 3: Start Wireshark Live Capture (1 min) . . . 3
Phase 4: Start eNodeB (1 min) . . . 3
Phase 5: Attach UEs (2 min) . . . 4
Phase 6: Send Emergency Alert (1 min) . . . 4
Phase 7: Verify Alert Reception . . . 5
Phase 8: Multi-Alert Test Sequence . . . 6
Troubleshooting Quick Reference . . . 6
Complete Startup Script . . . 7
Shutdown Procedure . . . 8

Manual Launch Guide — CMAS/PWS Emergency Alert Lab

Complete step-by-step procedure to launch the lab manually, from cold start to live Wireshark capture during alert broadcast.

Duration: ~8 minutes
Prerequisites: All hardware connected, SIM cards provisioned

Quick Reference

Component	Machine	Command
Core Network	PC1	`docker-compose up -d`
eNB	PC1	`sudo srsenb enb_pws.conf`
Wireshark	PC1	`wireshark -i any -k -f "sctp"`
Send Alert	PC1	`./send_alert.sh presidential "Test"`

Phase 1: Pre-flight Checks (2 min)

1.1 Verify Docker is Running

```
docker ps
# Should show: Running containers or empty list
```

1.2 Check bladeRF Device

```
bladeRF-cli -p
# Expected: Serial XXXXXXXXXX, bladeRF 2.0 (bladeRF xA4)

# If not detected:
sudo bladeRF-cli -L /usr/share/bladerf/fpga/hostedx40.rbf
```

1.3 Verify Config Files

```
ls -la /home/f2g/Desktop/CMAS-PWS/configs/srsran/
# Should see: enb_pws.conf, sib_pws.conf, etc.
```

1.4 Check Phone Settings

On each phone: - Settings → Safety & Emergency → Wireless Emergency Alerts - Enable: Presidential, Extreme, Severe, AMBER - For testing: Enable "Test alerts"

Phase 2: Start Core Network (2 min)

2.1 Start Docker Containers

```
cd /home/f2g/Telecom/Core-Network/openimss
docker-compose up -d
```

Wait 10 seconds for all services to initialize.

2.2 Verify Core Services

```
# Check containers running
docker ps --format "table {{.Names}}\t{{.Status}}" | head -10

# Check MME is listening
docker logs mme 2>&1 | tail -5
# Look for: "s1ap_server() [INFO]: Waiting for eNB connection"
```

2.3 Verify MME Ready for PWS

```
docker logs mme 2>&1 | grep -i "pws\|warning\|cbc"
# May show: PWS/CBS related initialization messages
```

Phase 3: Start Wireshark Live Capture (1 min)

3.1 Launch Wireshark

Open a new terminal:

```
# Option A: Capture all interfaces with SCTP filter
sudo wireshark -i any -k -f "sctp"

# Option B: Capture specific interface
sudo wireshark -i enp3s0 -k -f "sctp port 36412"

# Option C: CLI capture with tshark
sudo tshark -i any -f "sctp" -Y "s1ap"
```

3.2 Apply Display Filters

In Wireshark filter bar:

```
# All S1AP messages
s1ap

# Only WriteReplaceWarning
s1ap.procedureCode == 36

# CMAS SIB12
lte-rrc.systemInformationBlockType12_element

# ETWS SIB10/11
lte-rrc.systemInformationBlockType10_element || lte-rrc.systemInformationBlockType11_element
```

3.3 Key Messages to Watch

Message	Direction	Meaning
S1SetupRequest	eNB → MME	eNB connecting
S1SetupResponse	MME → eNB	Connection accepted
WriteReplaceWarningRequest	MME → eNB	Alert sent to eNB
WriteReplaceWarningResponse	eNB → MME	Alert acknowledged

Phase 4: Start eNodeB (1 min)

4.1 Launch srsENB

```
cd /home/f2g/Desktop/CMAS-PWS/configs/srsran
sudo srsenb enb_pws.conf
```

4.2 Expected Output

```
---  Software Radio Systems LTE eNodeB  ---
```

```
Opening 1 channels in RF device=bladeRF with args=
Set RX antenna to RX
...
==== eNodeB started ===
Type <t> to view trace
Setting frequency: DL=1830.0 MHz, UL=1735.0 MHz
Found bladeRF serial XXXXXXXXXX
```

4.3 Verify S1 Connection

In Wireshark, you should see: - `S1SetupRequest` from eNB - `S1SetupResponse` from MME

Check MME logs:

```
docker logs mme 2>&1 | grep -i "enb\|s1ap"
# Look for: "eNB-S1 accepted[172.22.0.1]"
# Look for: "Number of eNBs is now 1"
```

Phase 5: Attach UEs (2 min)

5.1 Power On Phones

1. Insert SIM cards with correct IMSI/K/OPC
2. Power on phones
3. Wait for LTE attachment (signal bars appear)

5.2 Verify Attachment

On eNB terminal, press `t` for trace:

```
------DL-------------------------------UL--------------------------------
rnti  cqi  ri  mcs  brate   ok  nok  (%)  pusch  pucch  phr  mcs  brate
  46   15   1   28   5.2M  100    2   2%   -8.0   -4.5   20   24   2.1M
```

5.3 Check Phone Connectivity

- Phone should show LTE symbol
- Try opening a webpage to verify data
- Check PLMN: Should show “00101” or “Test PLMN”

Phase 6: Send Emergency Alert (1 min)

6.1 Using Shell Script

```
cd /home/f2g/Desktop/CMAS-PWS/scripts

# Presidential Alert (cannot be disabled)
./send_alert.sh presidential "PRESIDENTIAL ALERT - This is a test"

# AMBER Alert
```

```
./send_alert.sh amber "AMBER Alert - Missing child in Yaoundé"

# Earthquake Alert (ETWS)
./send_alert.sh earthquake "EARTHQUAKE WARNING - Take cover immediately"

# Severe Weather Alert
./send_alert.sh severe "SEVERE WEATHER - Flash flood warning"
```

6.2 Using Python Script

```
python3 inject_write_replace_warning.py \
    --mme-ip 172.22.0.9 \
    --message-id 4370 \
    --message "EMERGENCY ALERT - This is a test"
```

6.3 Alert Types Reference

Command	Message ID	Description
presidential	4370	Cannot be disabled, highest priority
extreme	4371	Imminent extreme threat
severe	4373	Severe threat
amber	4379	Child abduction
earthquake	4352	ETWS earthquake
tsunami	4353	ETWS tsunami
test	4380	Monthly test

Phase 7: Verify Alert Reception

7.1 Phone Display

- Alert popup should appear on all attached phones
- Alert sound plays (if not silenced)
- Message content visible
- For Presidential: NO "Stop" option available

7.2 Wireshark Verification

Check for these messages in order:

#	Message	Filter
1	WriteReplaceWarningRequest	`s1ap.procedureCode == 36`
2	WriteReplaceWarningResponse	`s1ap.procedureCode == 36`
3	SIB12 broadcast	`lte-rrc.systemInformationBlockType12_element`

7.3 Check Message Content

In Wireshark, expand: - S1AP → WriteReplaceWarningRequest - Look for: `messageIdentifier`, `warningMessageContents`

7.4 Save Capture

File → Save As → `CMAS_test_YYYYMMDD_HHMM.pcapng`

Phase 8: Multi-Alert Test Sequence

8.1 Test Different Alert Types

```
# Wait 30 seconds between alerts for phones to clear

# Test 1: Presidential
./send_alert.sh presidential "Test 1 - Presidential Alert"
sleep 30

# Test 2: AMBER
./send_alert.sh amber "Test 2 - AMBER Alert"
sleep 30

# Test 3: Earthquake
./send_alert.sh earthquake "Test 3 - Earthquake ETWS"
sleep 30

# Test 4: Monthly Test
./send_alert.sh test "Test 4 - Monthly Required Test"
```

8.2 Verify Each Alert

For each alert, confirm: - [ ] Wireshark shows WriteReplaceWarningRequest - [ ] Phone displays alert popup - [ ] Correct message content - [ ] Appropriate alert tone

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Troubleshooting Quick Reference

Alert not appearing on phone

```
# Check SIB broadcast in Wireshark
Filter: lte-rrc.systemInformationBlockType12_element

# Check Message ID is in valid range
Valid CMAS: 4370-4399
Valid ETWS: 4352-4359

# Check phone settings
Settings → Safety → Emergency Alerts → Enable all
```

MME not forwarding alert

```
# Check MME logs
docker logs mme 2>&1 | grep -i "warning\|pws"

# Restart MME
docker restart mme
sleep 5

# Re-check
docker logs mme | tail -10
```

eNB not connected

```
# Check S1 connection
docker logs mme 2>&1 | grep "Number of eNBs"

# Restart eNB
# (Ctrl+C to stop, then restart)
sudo srsenb enb_pws.conf
```

No WriteReplaceWarningResponse

```
# eNB may not support PWS
# Check eNB configuration has SIB10/11/12 enabled
grep -i "sib1[012]" configs/srsran/sib_pws.conf
```

Complete Startup Script

Save as `start_cmas_lab.sh`:

```
#!/bin/bash
# Complete CMAS lab startup script

echo "=== Starting CMAS/PWS Lab ==="

# 1. Start Core
cd /home/f2g/Telecom/Core-Network/openimss
docker-compose up -d
sleep 10

# 2. Verify Core
echo ""
echo "=== Core Status ==="
docker ps --format "table {{.Names}}\t{{.Status}}" | grep -E "mme|hss"

# 3. Start Wireshark in background
echo ""
```

```
echo "=== Starting Wireshark ==="
sudo wireshark -i any -k -f "sctp" &
sleep 2

# 4. Start eNB
echo ""
echo "=== Starting eNB ==="
gnome-terminal --title="eNB PWS" -- bash -c "sudo srsenb /home/f2g/Desktop/CMAS-PWS/configs/srsra

echo ""
echo "=== Lab Started ==="
echo "1. Wait for 'eNB-S1 accepted' in MME logs"
echo "2. Power on phones"
echo "3. Run: ./scripts/send_alert.sh presidential \"Test message\""
```

Shutdown Procedure

```
# 1. Stop eNB (Ctrl+C in eNB terminal)

# 2. Stop Core
cd /home/f2g/Telecom/Core-Network/openimss
docker-compose down

# 3. Save Wireshark capture before closing
```

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